

Class 1 Reserved-Slot Strategies: Day-Level Formulation

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1 Introduction

1.1 Purpose and Reader Map

This note is a companion to `first_two_class_simulation_note_v2.tex`. It keeps the same day-level notation, the same patient behavior objects, and the same metric interpretation, but it adds two booking strategies in which Class 1 has protected daily capacity.

Core Policy Question

The pooled FCFS baseline asks which patient arrives first. The reserved-slot strategies ask whether some daily capacity should be protected for Class 1 before the two classes compete for the remaining slots.

The three strategies compared in this note are:

- **Pooled FCFS:** every class competes for the same S daily slots.
- **Strict Class 1 reservation:** $Q = 10$ slots per day are reserved for Class 1 only; unused reserved slots stay empty.
- **Class-1-first backfill reservation:** all Class 1 arrivals on a simulation day are handled before Class 2 arrivals; Class 1 searches reserved then general capacity on each residual day, and Class 2 may later backfill unused reserved slots.

1.1.1 What Changes from the FCFS Note

Object	FCFS baseline	Reserved-slot replacement
Daily capacity	one pooled capacity S	protected capacity Q plus general capacity $S - Q$
Calendar state	$X_{i,r}^D, N_r^D$	same $X_{i,r}^D$, plus audit-layer pool counts $X_{i,r,\text{res}}^D$ and $X_{i,r,\text{gen}}^D$
Offer rule	earliest residual day with $N_r^D < S$	earliest residual day with eligible pool capacity
No-offer logic	no pooled slot is available	no eligible pool is available for that class and policy
Service, cancellation, no-show	unchanged	unchanged

2 Model Formulation

2.1 Notation Inherited from v2

The simulation clock is still the day D . The basic objects are unchanged:

- S : total appointment slots available per day.
- H : rolling booking horizon in days.
- $\mathcal{I} = \{1, 2\}$: patient classes.
- $r \in \{0, 1, \dots, H - 1\}$: residual delay, where $r = 0$ means appointments scheduled for today.
- τ : original offered delay accepted by a booked patient.
- $X_{i,r}^D$: number of class- i patients scheduled for day $D + r$.
- $N_r^D = \sum_{i \in \mathcal{I}} X_{i,r}^D$: total scheduled patients on residual day r .

The same residual-delay/original-delay distinction remains:

Residual Delay vs. Original Delay

Residual delay r is part of the rolling calendar state and decreases when the day advances. Original delay τ is stored with the booking because balking and no-show behavior use the delay the patient was originally offered.

To make the policy formulas match the current implementation and remain compatible with the v2 notation, define the bookable residual-day set

$$\mathcal{R} \subseteq \{0, 1, \dots, H - 1\}.$$

In the current Python engine, same-day booking is allowed, so

$$\mathcal{R} = \{0, 1, \dots, H - 1\}.$$

If a future-only convention is desired, the same formulas apply after replacing $\mathcal{R}$ with $\{1, \dots, H - 1\}$.

2.2 Reservation Parameters

Let the protected class be

$$i^* = 1,$$

and let

$$Q = 10$$

be the number of Class 1 reserved slots per day. Since the baseline capacity is $S = 32$, the general pool has

$$S - Q = 22$$

slots per day.

The total day-level state $X_{i,r}^D$ is still the main state, but the reserved policies require an audit-layer pool label. Write

$$X_{i,r,\text{res}}^D = \text{number of class-}i \text{ patients scheduled in reserved slots on day } D + r,$$

and

$$X_{i,r,\text{gen}}^D = \text{number of class-}i \text{ patients scheduled in general slots on day } D + r.$$

Then

$$X_{i,r}^D = X_{i,r,\text{res}}^D + X_{i,r,\text{gen}}^D.$$

Define the pool occupancies

$$R_r^D = \sum_{i \in \mathcal{I}} X_{i,r,\text{res}}^D, \quad G_r^D = \sum_{i \in \mathcal{I}} X_{i,r,\text{gen}}^D,$$

so that

$$N_r^D = R_r^D + G_r^D.$$

The capacity constraints are

$$R_r^D \leq Q, \quad G_r^D \leq S - Q, \quad N_r^D \leq S.$$

Daily Capacity Split for the Implemented Strategy

Pool	Capacity	Eligibility
Reserved pool	$Q = 10$	Class 1 always; Class 2 only under the Class-1-first backfill policy after that day's Class 1 arrivals have been handled
General pool	$S - Q = 22$	Class 1 overflow and Class 2
Total day	$S = 32$	all scheduled patients, regardless of pool label

2.3 Patient Behavior is Unchanged

The class behavior object remains

$$C_i = \{\lambda_i, b_i(\tau), \phi_i, \xi_i(\tau)\}.$$

The reservation policies do not change arrival rates, balking, cancellation, or no-show behavior. They only change which residual day and which capacity pool an arriving patient is offered.

What the Reservation Strategy Does Not Change

For a class- i patient offered delay τ , the balking probability remains $b_i(\tau)$. For a booked class- i patient scheduled in the future, the daily cancellation probability remains ϕ_i . At service time, the no-show probability remains $\xi_i(\tau)$.

3 Booking Rules

3.1 Pooled FCFS Baseline

The pooled FCFS baseline searches the residual days in order and offers the first day with open total capacity:

$$\tau_{\text{FCFS}} = \min \{r \in \mathcal{R} : N_r^D < S\}.$$

If the set is empty, the patient receives no offer.

If the patient accepts, the simulator increments

$$X_{i, \tau_{\text{FCFS}}}^D \leftarrow X_{i, \tau_{\text{FCFS}}}^D + 1$$

and stores the audit record

$$(i, \tau_{\text{FCFS}}).$$

FCFS in One Line

Under pooled FCFS, class labels do not affect the offer rule. They affect behavior after the offer through $b_i(\tau)$, ϕ_i , and $\xi_i(\tau)$, but not the capacity search itself.

3.2 Strict Class 1 Reservation

Under strict reservation, Class 1 has first access to the reserved pool on each day. If the reserved pool is already full on the earliest eligible day, Class 1 can compete for general slots. Class 2 can use only the general pool. Unused reserved slots stay empty and are carried forward as protected future capacity until the appointment day arrives.

3.2.1 Class 1 Offer Rule

For a Class 1 arrival, the strict policy searches residual days in increasing r . On each residual day, it checks the reserved pool first and the general pool second:

$$\tau_{1, \text{strict}} = \min \{r \in \mathcal{R} : R_r^D < Q \text{ or } G_r^D < S - Q\}.$$

The pool label is

$$z_{1,\text{strict}}(r) = \begin{cases} \text{res}, & R_r^D < Q, \\ \text{gen}, & R_r^D = Q \text{ and } G_r^D < S - Q. \end{cases}$$

If both pools are unavailable for every $r \in \mathcal{R}$, the Class 1 patient receives no offer. If the patient accepts, the corresponding pool count is incremented:

$$X_{1,\tau,\text{res}}^D \leftarrow X_{1,\tau,\text{res}}^D + 1 \quad \text{or} \quad X_{1,\tau,\text{gen}}^D \leftarrow X_{1,\tau,\text{gen}}^D + 1.$$

3.2.2 Class 2 Offer Rule

For a Class 2 arrival under strict reservation, the reserved pool is invisible:

$$\tau_{2,\text{strict}} = \min \{r \in \mathcal{R} : G_r^D < S - Q\}.$$

If the set is empty, the Class 2 patient receives no offer even if some reserved capacity remains unused:

$$R_r^D < Q \quad \text{does not help Class 2 under strict reservation.}$$

If the patient accepts, the simulator increments

$$X_{2,\tau,\text{gen}}^D \leftarrow X_{2,\tau,\text{gen}}^D + 1.$$

Strict Reservation Interpretation

Strict reservation protects Class 1 access at the cost of possible empty protected capacity. It is not a utilization-maximizing policy; it is an access-protection policy.

3.3 Class-1-First Backfill Reservation

The Class-1-first backfill policy keeps the same $Q = 10$ protected slots, but it changes the within-day order of the booking step. Arrivals on day D are handled in two priority stages. First, all Class 1 arrivals are offered appointments. Second, Class 2 arrivals are offered appointments. Class 2 cannot use protected capacity until the Class 1 stage for that simulation day is complete.

3.3.1 Stage 1: Class 1 Arrivals

Let the randomized arrival sequence on day D be

$$A^D = (a_1^D, a_2^D, \dots, a_{MD}^D),$$

where each $a_k^D \in \mathcal{I}$. In the first stage, the policy processes only entries with

$$a_k^D = 1.$$

For each Class 1 arrival, residual days are searched chronologically. Within each residual day, reserved capacity is checked before general capacity:

$$\tau_{1,\text{bf}} = \min \{r \in \mathcal{R} : R_r^D < Q \text{ or } G_r^D < S - Q\}.$$

The pool label is

$$z_{1,\text{bf}}(r) = \begin{cases} \text{res}, & R_r^D < Q, \\ \text{gen}, & R_r^D = Q \text{ and } G_r^D < S - Q. \end{cases}$$

Thus the Class 1 search order is

$$(0, \text{res}), (0, \text{gen}), (1, \text{res}), (1, \text{gen}), \dots$$

If both pools are unavailable for every $r \in \mathcal{R}$, the Class 1 patient receives no offer.

No Horizon-Wide Reserved-Only Pass

Class-1-first backfill reservation does not search all future reserved slots before general slots. If day 0 has no reserved capacity but has a general slot, and day 1 has a reserved slot, a Class 1 patient is offered the day 0 general slot.

3.3.2 Stage 2: Class 2 Arrivals

After all Class 1 arrivals from day D have been handled, the policy processes the Class 2 arrivals from that same simulation day.

Class 2 first searches the general pool and then may use leftover reserved capacity:

$$\tau_{2,\text{bf}} = \min \{ r \in \mathcal{R} : G_r^D < S - Q \text{ or } R_r^D < Q \}.$$

The pool label for Class 2 is

$$z_{2,\text{bf}}(r) = \begin{cases} \text{gen}, & G_r^D < S - Q, \\ \text{res}, & G_r^D = S - Q \text{ and } R_r^D < Q. \end{cases}$$

Thus the Class 2 search order is

$$(0, \text{gen}), (0, \text{res}), (1, \text{gen}), (1, \text{res}), \dots$$

where a reserved slot is available to Class 2 only because the Class 1 stage for day D has already ended.

Class-1-First Backfill Interpretation

Class-1-first backfill reservation is a compromise. Class 1 receives priority access to Q protected daily slots, but unused protected capacity can still be converted into Class 2 access after the day-level Class 1 stage.

4 Visual Calendar Logic

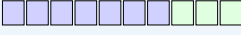
4.1 One Residual Day

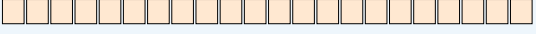
The reservation logic is easiest to see on a single residual day r . The day has $Q = 10$ protected slots and $S - Q = 22$ general slots.

Pool Occupancy on One Residual Day

Suppose $S = 32$, $Q = 10$, and the post-cancellation state for one residual day is

$$R_r^D = 7, \quad G_r^D = 22, \quad N_r^D = 29.$$

reserved $Q = 10$  $R_r^D = 7$, three protected slots open

general $S - Q = 22$  $G_r^D = 22$, no general slots open

Under strict reservation, a Class 1 patient can be offered this day, but a Class 2 patient cannot. Under Class-1-first backfill reservation after the day-level Class 1 stage, a Class 2 patient can use one of the three open protected slots.

4.2 Residual-Day Search

Comparing the Earliest Feasible Day

Take $H = 4$, $S = 32$, $Q = 10$, and the post-cancellation state below.

Count	day D $r = 0$	day $D + 1$ $r = 1$	day $D + 2$ $r = 2$	day $D + 3$ $r = 3$
R_r^D	10	8	10	6
G_r^D	22	22	20	13
N_r^D	32	30	30	19

The implied offers are:

Policy and class	Earliest offer	Reason
Pooled FCFS, either class	$r = 1$	$N_1^D = 30 < 32$
Strict reservation, Class 1	$r = 1$, reserved	$R_1^D = 8 < 10$
Strict reservation, Class 2	$r = 2$, general	$G_1^D = 22$ is full, but $G_2^D = 20 < 22$
Class-1-first backfill reservation, Class 2 after Class 1 stage	$r = 1$, reserved if unused	leftover protected slots can be backfilled

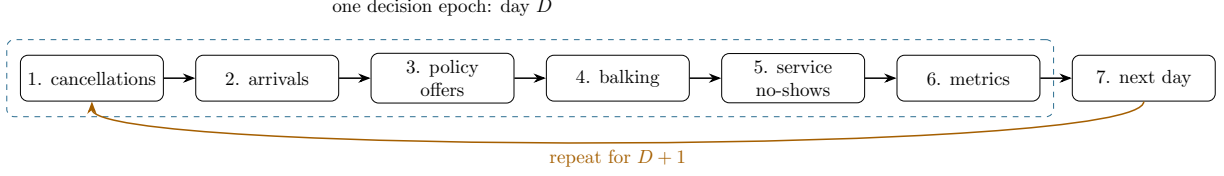
5 Daily Simulation Mechanics

5.1 Unchanged Daily Order

The daily event order remains the same as in the v2 note:

1. **Compute cancellations for the day.** Future appointments independently cancel with probability ϕ_i .

2. **Simulate daily arrivals.** $M_i^D \sim \text{Poisson}(\lambda_i)$.
3. **Generate a randomized within-day arrival order.**
4. **Apply the configured offer rule.** This is the only step changed by the reserved-slot strategies.
5. **Simulate balking.** A class- i patient offered delay τ accepts with probability $1 - b_i(\tau)$.
6. **Simulate no-shows and service for day D .**
7. **Compute metrics and advance to day $D + 1$.**



5.2 Strict Reservation Day

Under strict reservation, the randomized arrival order is processed once. The difference from pooled FCFS is that the feasible capacity set depends on the patient's class:

$$\mathcal{F}_{1,\text{strict}}^D = \{(r, \text{res}) : r \in \mathcal{R}, R_r^D < Q\} \cup \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\},$$

where reserved capacity is checked before general capacity on the same residual day. For Class 2,

$$\mathcal{F}_{2,\text{strict}}^D = \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\}.$$

If $\mathcal{F}_{i,\text{strict}}^D$ is empty, that patient receives no offer. The simulator then continues to later arrivals because a later Class 1 arrival may still have protected capacity even when a Class 2 arrival has no general capacity.

5.3 Class-1-First Backfill Reservation Day

Under Class-1-first backfill reservation, the day has two priority stages.

1. **Class 1 stage.** Class 1 arrivals search residual days chronologically and check reserved capacity before general capacity on each residual day:

$$(0, \text{res}), (0, \text{gen}), (1, \text{res}), (1, \text{gen}), \dots$$

Equivalently,

$$\mathcal{F}_{1,\text{bf}}^D = \{(r, \text{res}) : r \in \mathcal{R}, R_r^D < Q\} \cup \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\},$$

with the ordered search above. This is not a horizon-wide reserved-only pass.

2. **Class 2 stage.** After all Class 1 arrivals have been handled, Class 2 arrivals search general capacity first and then leftover protected capacity on each residual day:

$$(0, \text{gen}), (0, \text{res}), (1, \text{gen}), (1, \text{res}), \dots$$

Equivalently,

$$\mathcal{F}_{2,\text{bf}}^D = \{(r, \text{gen}) : r \in \mathcal{R}, G_r^D < S - Q\} \cup \{(r, \text{res}) : r \in \mathcal{R}, R_r^D < Q\},$$

with the ordered search above.

This staged rule is the mathematical version of “Class 1 gets priority within the day; if protected slots remain after that Class 1 stage, Class 2 may fill them later.”

6 Comparison to Pooled FCFS

6.1 Policy Effects

Feature	Pooled FCFS	Strict Class 1 reservation	Class-1-first backfill reservation
Capacity structure	all S slots pooled	Q protected, $S - Q$ general	Q protected first, then releasable
Class 1 access	competes only by arrival order	protected first, then general overflow	handled before Class 2; protected then general on each residual day
Class 2 access	can use any open slot	can use only general slots	can use general slots and unused protected slots after the Class 1 stage
Unused reserved slots	not applicable	may remain empty	can be backfilled by Class 2
Utilization tendency	highest when demand is sufficient and no-shows are low	can fall if Class 1 demand is not enough to use Q	closer to pooled FCFS because unused protection can be recovered
Equity/access tendency	neutral in the offer rule; differences come from behavior	shifts access toward Class 1	shifts access toward Class 1, but less rigidly

6.2 Expected Metric Patterns

The main metrics are unchanged:

$$\rho^{\text{attended}} = \frac{\text{served slots}}{S \times \text{measured days}}, \quad \text{served rate}_i = \frac{\text{served}_i}{\text{arrivals}_i}.$$

The reservation strategies mainly affect the path to those metrics:

$$\text{offer rule} \rightarrow \tau \rightarrow b_i(\tau) \rightarrow \text{booked patients} \rightarrow \phi_i, \xi_i(\tau) \rightarrow \text{served patients}.$$

Interpretation of the Comparison

Strict reservation can improve Class 1 access while lowering aggregate utilization if protected slots are not filled. Class-1-first backfill reservation tests whether most of the Class 1 access protection can be kept while reducing the utilization loss by letting Class 2 backfill unused protected capacity.

6.3 No-Offer Interpretation

The no-offer event has a slightly different meaning across the strategies:

Policy	No-offer meaning
Pooled FCFS	no day in $\mathcal{R}$ satisfies $N_r^D < S$
Strict reservation, Class 1	no day has either $R_r^D < Q$ or $G_r^D < S - Q$
Strict reservation, Class 2	no day has $G_r^D < S - Q$, even if $R_r^D < Q$ somewhere
Class-1-first backfill reservation, Class 1	no day has either $R_r^D < Q$ or $G_r^D < S - Q$ during the Class 1 stage
Class-1-first backfill reservation, Class 2	no general slot and no leftover protected slot are available after the Class 1 stage

7 Implementation Mapping

7.1 Config Objects

The new implementation fields map directly to the notation:

Code field	Notation	Meaning
<code>slots_per_day</code>	S	total daily capacity
<code>reserved_class_id</code>	i^*	protected class; here $i^* = 1$
<code>reserved_slots_per_day</code>	Q	protected daily capacity; here $Q = 10$
<code>release_reserved_slots</code>	backfill flag	false for strict reservation, true for Class-1-first backfill reservation
<code>Booking.reserved_slot</code>	pool label z	true means the appointment was placed in the reserved pool

7.2 Code Path

The policy logic appears in the booking step:

- `find_earliest_open_day` computes the earliest feasible (r, z) pair.
- `process_daily_arrivals` applies either the one-pass rule or the Class-1-first backfill priority-stage rule.
- In the backfill rule, all Class 1 arrivals are processed before non-reserved classes.
- For Class 1, `find_earliest_open_day` searches each residual day as reserved then general.

- For Class 2 in the backfill stage, the search order is general then leftover reserved on each residual day.

Review Against the Implemented Reserved-Slot Model

- The original day-level state notation $X_{i,r}^D$, N_r^D , r , τ , S , and H is unchanged.
- The new audit-layer pool counts R_r^D and G_r^D split the same daily capacity into reserved and general pools.
- $Q = 10$ protected slots are reserved for Class 1 out of $S = 32$.
- In strict reservation, unused protected slots stay unavailable to Class 2.
- In Class-1-first backfill reservation, Class 1 gets the day-level priority stage and Class 2 can backfill unused protected capacity afterward.
- Cancellation, no-show, service, and metric definitions remain the same as in the FCFS v2 note.

8 Conclusion

The reservation strategies are not separate patient behavior models. They are changes to the offer rule inside the same day-level appointment simulator. Pooled FCFS uses one capacity constraint

$$N_r^D < S.$$

Strict Class 1 reservation replaces that with class-dependent eligibility over

$$R_r^D < Q \quad \text{and} \quad G_r^D < S - Q.$$

Class-1-first backfill reservation keeps a protected day-level priority stage for Class 1, then relaxes the strict rule by letting Class 2 use protected capacity that Class 1 did not reserve.

The comparison should therefore be read as an access/utilization tradeoff:

$$\text{more Class 1 protection} \quad \Longleftrightarrow \quad \text{less automatic pooling of daily capacity.}$$